BUILDING ADVANCED AUTONOMOUS AI WITH SAPPHIRE

Architectural Blueprints, Neural Training, Memory, Planning, Tools & Autonomy

1. Autonomous AI Architecture Principles

An **Autonomous AI Agent** in Sapphire operates continuously in an environment, senses system telemetry, formulates plans, trains neural policy models, executes system actions via registered tools, evaluates feedback, and self-corrects.

```
// Sapphire Complete AI Agent Pipeline (.sp)
fn run_pipeline() {
    let ds = ml.dataset.random(1000, 16, 2);
    let model = ml.model.mlp([16, 64, 2], "relu");
    let result = ml.train.fit(model, ds, ml.loss.mse, ml.optim.adam(0.01), 3, 32, 2);

    let ai_eval = ai.prompt("Final loss is {result.final_loss}. Is model ready?");
    agent.memory.remember("model_loss", result.final_loss);

    let goal = "Deploy model and notify admin";
    let agent_report = agent.autonomy.run_loop(goal, 4);
    print("Agent status: {agent_report['finished']}");
}

run_pipeline();
```